

Sujal Rajbhandari

AI/ML Engineer | Generative AI · RAG · LangGraph · MCP

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PERSONAL PROFILE

AI/ML Engineer with 2 years of hands-on experience designing, building, and deploying production-grade generative AI and machine learning systems. Currently serving as the sole machine learning engineer at Resimator Oy, where I have independently led the AI/ML development of 10+ production systems spanning Generative AI, RAG pipelines, LangGraph multi-agent workflows, MCP server architectures, LLM-powered document extraction, recommendation systems, anomaly detection, and forecasting. Strong end-to-end ownership across the ML lifecycle from problem framing and prototyping through deployment on AWS using FastAPI, Docker, Lambda, EC2, S3, ALB, and Redis. Passionate about applied generative AI, building reliable LLM systems, and delivering measurable business value through AI.

KEY ACHIEVEMENTS

- **Sole ML Engineer at Resimator Oy** independently led the AI/ML architecture, development, and deployment of 10+ production AI systems serving real business use cases.
- **Built end-to-end LLM-powered document extraction systems** (invoices and legal lease agreements) that convert unstructured PDFs and scanned images into validated structured JSON, significantly reducing manual data entry.
- **Designed and deployed an MCP server architecture** powering an AI business analytics chatbot capable of answering complex natural-language business questions with KPIs, trends, insights, and chart-ready outputs.
- **Implemented RAG pipelines and LangGraph-based multi-agent workflows** for grounded, context-aware AI applications using vector databases (Pinecone, Chroma, FAISS, Qdrant) and embedding models.

PROFESSIONAL EXPERIENCE

Resimator Oy Jan 2025 – June 2026

AI/ML Engineer (Sole Machine Learning Engineer) Remote

- **Architected and deployed an AI-powered invoice extraction system** using LLMs, OCR pipelines, FastAPI, and AWS to extract structured invoice data from unstructured PDFs and images, reducing manual data entry workload.
- **Built a legal lease agreement information extraction system** using LLM-powered NLP pipelines to parse complex legal documents and extract parties, dates, monetary terms, key clauses, and contractual obligations into structured JSON outputs.
- **Designed prompt engineering and validation pipelines** to ensure reliable LLM-generated JSON outputs, using structured schemas and Pydantic-based validation logic for production-ready document extraction workflows.
- **Integrated Gemini and OpenAI APIs** into AI applications for document understanding, chatbot responses, information extraction, reasoning, and business insight generation.
- **Engineered multi-agent AI workflows using LangGraph**, enabling stateful, graph-based orchestration of AI agents for tool usage, reasoning, and multi-step task automation.
- **Designed and developed an MCP server architecture** that enables AI agents and chatbots to access external tools, APIs, business data, and contextual information for dynamic, data-aware responses.
- **Built an AI business analytics chatbot** on top of the MCP server, allowing users to ask simple and complex business questions and receive answers with KPIs, trends, insights, and auto-generated chart recommendations.
- **Developed RAG pipelines** using LangChain, vector databases (Pinecone, Chroma, FAISS, Qdrant), and embedding models to support document-grounded question answering with reduced hallucination and improved contextual accuracy.
- **Built recommendation systems** combining collaborative filtering, content-based filtering, hybrid scoring, and embedding-based similarity. Evaluated performance using Precision@K, Recall@K, and NDCG@K.
- **Developed an anomaly detection and reasoning system** that identifies irregular business or operational patterns and generates interpretable LLM-based explanations for each flagged anomaly.
- **Built additional AI systems** including an image generation chatbot, a menu scraping pipeline using web scraping and NLP, and forecasting models for short-term and medium-term business prediction.
- **Deployed multiple AI systems on AWS** using EC2, S3, Lambda, ALB, Redis, Docker, REST APIs, and FastAPI, owning model serving and deployment lifecycle end-to-end.

Creatu Developers Dec 2022 – Mar 2023

Data Science Intern Kathmandu, Nepal

- Performed exploratory data analysis, data cleaning, and feature preparation on real-world datasets using Pandas, NumPy, Matplotlib, Seaborn, and Plotly.
- Built time-series prediction models for business trend analysis and short-term forecasting using statistical and machine learning approaches.
- Developed interactive visualizations and analytical dashboards to communicate insights and patterns to stakeholders.

SELECTED PROJECTS

RAG-Powered AI Chatbot with LangGraph & Vector Databases | *Python, LangGraph, LangChain, Pinecone, Chroma, FAISS, Qdrant, OpenAI, Gemini, FastAPI*

Designed and developed a retrieval-augmented generation chatbot with LangGraph-based multi-agent orchestration and vector database integration. Retrieves relevant context from documents and business data to generate grounded, context-aware responses with reduced hallucination.

MCP-Based AI Business Analytics Chatbot | *Python, MCP, LangGraph, LangChain, OpenAI, Gemini, FastAPI, REST APIs*

Built a chatbot using the MCP server architecture to fetch business data dynamically and answer user queries in natural language. Supports complex analytical questions, providing KPIs, business insights, trend summaries, and chart-ready outputs.

AI-Powered Invoice Extraction System | *Python, Gemini, OpenAI, Tesseract OCR, Pydantic, FastAPI, AWS*

Engineered an end-to-end LLM-based invoice extraction system that processes unstructured PDFs and images, extracts invoice fields and line items, validates structured JSON outputs, and exposes results through REST API endpoints.

Legal Lease Agreement Information Extraction | *Python, Gemini, NLP, OCR, pdf2image, PyMuPDF, FastAPI*

Built a legal document extraction system for lease agreements, including scanned and image-based PDFs. Extracts clauses, parties, dates, rent details, deposit amounts, termination clauses, and obligations into structured outputs for legal-tech workflows.

Hybrid Recommendation System | *Python, Scikit-learn, TensorFlow, Embeddings*

Built a hybrid recommendation system combining collaborative filtering, content-based filtering, popularity ranking, and embedding-based similarity. Evaluated with Precision@K, Recall@K, and NDCG@K.

Anomaly Detection & Reasoning System | *Python, Scikit-learn, Pandas, LLMs*

Designed an anomaly detection pipeline that combines statistical and ML-based detection methods with LLM-generated reasoning, allowing users to understand why unusual patterns were flagged.

TECHNICAL SKILLS

Languages: Python, SQL

Generative AI & LLMs: LangChain, LangGraph, RAG Pipelines, MCP, Multi-Agent Systems, OpenAI API, Gemini API, Prompt Engineering, Structured JSON Extraction

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, PyTorch, Hugging Face Transformers, Recommendation Systems, Time Series Forecasting, Anomaly Detection

Vector Databases: Pinecone, Chroma, FAISS, Qdrant

Document AI & OCR: Tesseract OCR, pdf2image, PyMuPDF, Pydantic, JSON Validation, Document Parsing

Cloud & MLOps: AWS (EC2, S3, Lambda, ALB), Redis, Docker, FastAPI, REST APIs

Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Plotly, EDA, Data Cleaning, KPI Analysis, Data Visualization

Databases: MySQL, PostgreSQL, SQL

Tools: Git, GitHub, Bitbucket, Jupyter Notebook, Google Colab, VS Code, Postman, ClickUp

EDUCATION

Bachelor of Science in Information Technology (BSIT) 2023 – Present

Concentration: Artificial Intelligence & Machine Learning Kathmandu, Nepal

- Current GPA: 3.85 / 4.0
- Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming, Software Engineering, Web Technologies, Operating Systems, Computer Networks, Calculus, Statistics and Probability.

COURSES & CERTIFICATIONS

- Data Science Certificate
- Unsupervised Learning, Recommenders & Reinforcement Learning
- MCP with AI Agents
- Vector Databases with RAG
- Developing Generative AI Applications

LANGUAGES

English (Professional Working Proficiency) • **Nepali** (Native)